Practice Problems
Calculus Physics II

Name: _____ **Date:** _____

Chapter 54 – EM Spectrum

1) Most of the waves in the electromagnetic spectrum are

   A) red light.
   B) blue light.
   C) green light.
   D) Invisible to the human eye.

2) Which one of the following lists gives the correct order of the electromagnetic spectrum from *low to high frequencies*?

   A) radio waves, infrared, microwaves, ultraviolet, visible, x-rays, gamma rays
   B) radio waves, ultraviolet, x-rays, microwaves, infrared, visible, gamma rays
   C) radio waves, microwaves, infrared, visible, ultraviolet, x-rays, gamma rays
   D) radio waves, microwaves, visible, x-rays, infrared, ultraviolet, gamma rays
   E) radio waves, infrared, x-rays, microwaves, ultraviolet, visible, gamma rays

3) Which of these electromagnetic waves have the shortest wavelength?

   A) Visible light waves
   B) infrared waves
   C) ultraviolet waves
   D) X-rays
   E) radio waves

4) Compared to ultraviolet waves, the frequency of infrared waves is

   A) lower.
   B) the same.
   C) higher.

5) A main difference between a radio wave and a visible light EM wave is

   A) wavelength.
   B) frequency.
   C) both of these
   D) neither of these

6) Sunburns are produced by

   A) infrared light.
   B) visible light.
   C) ultraviolet light.
   D) all of the above
   E) none of the above

7) What EM wave is emitted from the nucleus of the atom?

A) infrared light.
B) visible light.
C) ultraviolet light.
D) all of the above
E) none of the above

8) What EM wave that are created by acceleration of electric charge?

A) radio waves.
B) gamma rays.
C) ultraviolet light.
D) all of the above
E) none of the above

9) What EM wave is emitted from electron transitions in atom?

A) infrared light.
B) visible light.
C) gamma rays.
D) all of the above
E) none of the above

10) Which part of the EM spectrum is used determine if a person has a fever?

A) Ultraviolet.
B) X-ray.
C) infrared light.
D) visible light.
E) Microwave.

11) A radio wave antenna made with a long straight metal rod detects

A) the electric field
B) the magnetic field
C) both the magnetic and electric field
D) neither the magnetic or electric field.

12) Radio waves that are transmitted using FM modulate the

A) Frequency.
B) Amplitude.
C) Zeros and ones.

13) Microwave radiation has a shorter wavelength than

A) Visible light.
B) x-rays.
C) Ultraviolet light.
D) Radio waves.
E) All of the above.

14) Cosmic Microwave Background is EM waves from

A) a super nova.
B) the big bang.
C) a neutron star.
D) the center of the Milky Way galaxy.
E) microwave ovens.

15) Infrared radiation is emitted by

A) Anything that has thermal energy.
B) The nucleus of the atom.
C) Electron transition in the atom.
D) Accelerating electrons.

16) For visible light, which color has the shortest wavelength?

A) Green
B) Red
C) Yellow
D) Blue
E) Orange
F) Violet

17) What is the Photopic part of the EM spectrum?

A) All of the EM spectrum.
B) All the spectrum above Ultraviolet light.
C) All the spectrum below visible light.
D) The part of the spectrum that humans can see.
E) The part of the spectrum that is used in medicine.

18) Which part of the Ultraviolet spectrum is Ionizing radiation?

A) UVA
B) UVB
C) UVC
D) All of the above.

19) What does Ionizing radiation do to atoms?

A) Adds charge to the atom.
B) Takes charge away from the atom.
C) Strips electrons from the atom.
D) Both A and B
E) None of the above.

20) Which type of x-ray is used for medical applications?

A) Soft x-rays
B) Hard x-rays
C) All x-rays

KEY

1) D
2) C
3) D
4) A
5) C
6) C
7) E
8) A
9) B
10) C
11) A
12) A
13) D
14) B
15) A
16) F
17) D
18) C
19) C
20) B